CPNT 217 - Introduction to Network Systems

Description
This course will introduce students to the fundamentals of network systems. Students will learn about networking basics and architecture, networking and software development, networking protocols, troubleshooting, and configuring virtual machines, Windows and Linux servers. Students will also learn about the functions of Active Directory in servers and performing socket programming with scripting language.

3 Credits

Time Guidelines
The standard instructional time for this course is 75 hours.

Course Assessment

Quizzes	10%
Labs	60%
Exams	30%
Total:	100%

Use of Turnitin

The instructor may submit student work in this course to Turnitin's text-matching software program, to help assess the academic integrity of student work in this course. Turnitin results may be considered as one piece of evidence in academic misconduct hearings. Turnitin is an American company that keeps no unencrypted student identity data in the United States unless the students themselves choose to share this information in their submission. Students should limit their sharing of personally identifiable information by not including their names and student ID numbers within the text body of submitted assignments. SAIT has carefully reviewed this company's data management procedures.

Other Course Information
Learner Engagement:

In order to be successful, the learner is expected to be engaged in learning activities for a total of 9 to 12 learning hours per course per week, which includes both in-class and out-of-class time.

Course Learning Outcomes
1. Summarize network architecture principles.

Objectives:

1.1 Describe how basic networks operate.

1.2 Identify hardware used in building a computer network.

1.3 Outline functions of computer networking hardware.

1.4 Summarize considerations for designing network topologies.

1.5 Review common network topologies for different use cases.

1.6 Explain the impact of cable selection on effective networking.

2. Relate networking to software deployment.

Objectives:

2.1 Relate software functionality to networking.

2.2 Describe the role of software in networking operations.

2.3 Summarize the functions of an Application Programming Interface (API).

2.4 Associate APIs with software development and networking.

2.5 Summarize the functions of scripts in software development.

2.6 Associate scripts with software development and networking.

3. Analyze the functions of networking protocols.

Objectives:

3.1 Define the term networking protocols.

3.2 Outline the different types of network protocols.

3.3 Relate protocols to networking reference models (TCP-IP and OSI).

3.4 Outline the functions of TCP-IP.

3.5 Outline the functions of OSI.

3.6 Describe how TCP-IP is used in networking.

3.7 Describe how OSI is used in networking.

3.8 Select a networking protocol based on required functionality.

4. Explain industry standard troubleshooting methodology.

Objectives:

4.1 Explain the importance of troubleshooting.

4.2 Review troubleshooting methodologies.

4.3 Identify commonalities between software development and networking troubleshooting.

4.4 Describe the role of networking utilities for troubleshooting.

4.5 Apply a troubleshooting process to a given scenario.

5. Demonstrate proper configuration of virtual machines.

Objectives:

5.1 Explain virtualization concepts regarding hardware.

5.2 Explain virtualization concepts regarding software.

5.3 Explain how cloud computing works.

5.4 Associate cloud computing and virtualization in industry.

5.5 Outline common uses of cloud computing and virtualization in industry.

5.6 Describe applications of cloud computing in software development.

5.7 Demonstrate the configuration of virtualization software.

5.8 Use virtualization software to create virtual machines.

6. Perform basic configuration of Windows Server.

Objectives:

6.1 Explain the role of servers in networks.

6.2 Demonstrate installation of Windows Server OS.

6.3 Identify services required from a server.

6.4 Demonstrate configuration of Windows Server OS based on requirements.

7. Explain the functions of Active Directory in servers.

Objectives:

7.1 Describe Active Directory (AD) as a software service.

7.2 Outline common uses of AD in industry.

7.3 Define AD policy.

7.4 Describe how AD policies can influence software functionality.

8. Perform basic configuration of Linux Server.

Objectives:

8.1 Compare Windows OS and Linux OS.

8.2 Demonstrate installation of Linux Server OS.

8.3 Demonstrate configuration of Linux OS.

9. Perform socket programming with an interpreted scripting language.

Objectives:

9.1 Explain the components of a socket.

9.2 Outline common uses of sockets in networking.

9.3 Outline common uses of sockets in application creation.

9.4 Describe common issues in socket use cases.

9.5 Describe solutions to common socket issues.

9.6 Use an interpreted scripting language to program a socket.

SAIT Policies and Procedures:
Students should familiarize themselves with SAIT's Policies and Procedures, including but not limited to

those in the Academic Student, Conduct, and Health, Safety, and Environment domains.

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